
Physics-Gated Correction Discovery: Why Constraining the Search Space Beats Scaling It in Noisy Symbolic Regression

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Abstract

Symbolic regression (SR) methods such as PySR, AI Feynman, and PhySO discover equations *from scratch*, an approach that overfits on noisy physical data: under matched residuals, PySR’s structural accuracy is non-monotonic in noise and does not improve when its search budget is doubled. We argue that scientific discovery is rarely *tabula rasa*—it is *correction-first*: given a known classical law and anomalous data, the real task is to find the minimal physically-valid correction Δ . We present **Anomaly-Driven Correction Discovery (ADCD)**, which restricts search to a correction subspace and cascades three physical filters—AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency (ARC)—*before* a JAX optimizer fits any parameter, with Bayesian Information Criterion (BIC) reranking. On a 9-scenario benchmark (mean $80.4\% \pm 7.4\%$ across 16 seeds; 88.9% at the disclosed reference seed), ADCD reaches 88.9% at 5% noise versus 11.1% for PySR—a *seed-independent* gap: ADCD’s *worst* of 16 seeds still beats PySR with doubled budget. Removing all gates drops recovery by 44.5 pp. ADCD is orthogonal to—not competitive with—existing SR: it restricts the problem they solve to a physically-motivated subspace where they fail under noise.

1 Introduction

The dominant paradigm in symbolic regression (SR) is *tabula rasa* discovery: take raw (X, y) and evolve an algebraic expression $f(X) \approx y$ from primitive operators. Pioneered by Eureqa Schmidt & Lipson [2009], AI Feynman Udrescu & Tegmark [2020], and PySR Cranmer [2020], and extended by deep RL approaches such as DSR Petersen et al. [2021] and unit-constrained PhySO Tenachi et al. [2023], these systems recover compact laws in *low-noise* regimes. Under realistic observational noise, however, their structural accuracy degrades and is often *non-monotonic* in noise: a larger candidate pool does not reliably translate into the correct functional *class* (Section 3).

We observe that this difficulty is partly one of *problem formulation*. Science rarely discovers from a blank slate—it *corrects*. From the relativistic kinetic-energy expansion $\frac{1}{2}mv^2(1 + \frac{3}{4}v^2/c^2 + \dots)$, to Debye-screened Coulomb potentials $V(r) \propto r^{-1}e^{-r/\lambda_D}$, to turbulent drag $F = bv + \alpha v^2$, progress takes a structurally-correct classical law and appends a *minimal correction* Δ that reconciles it with anomalous data. This restricts the hypothesis space from the full function space $\mathcal{F}$ to a correction subspace $\mathcal{D} \subset \mathcal{F}$ —an exponential reduction informed by the known physics.

We propose **Anomaly-Driven Correction Discovery (ADCD)**, built on three contributions:

C1 Correction-first paradigm. Search for Δ in $\mathcal{D}$, not f in $\mathcal{F}$, given a classical law $y_{\text{classical}} = g(X; \phi)$ and anomalous observations.

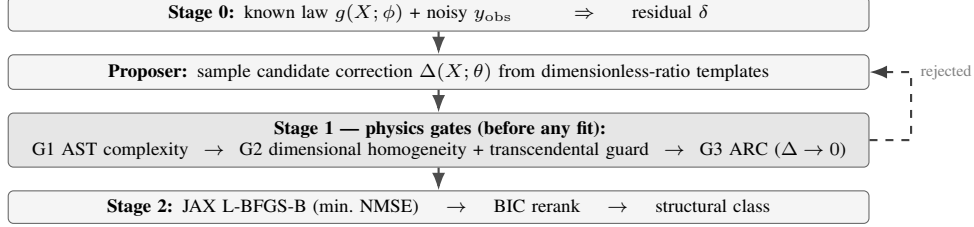


Figure 1: ADCD pipeline. Physics gates (G1–G3) screen candidates *before* any parameter is fit; rejected candidates are re-sampled.

C2 Physics-gated cascade before optimization. Three filters—AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency (ARC, $\Delta \rightarrow 0$ at the classical boundary)—screen candidates *before* any parameter is fit.

C3 Categorical immunity to trigonometric overfitting. Because dimensional/ARC gates reject nested $\sin(\sin(\dots))$ that dominate unconstrained Pareto fronts at 0% noise, ADCD’s accuracy is stable where PySR’s is non-monotonic.

We deliberately scope this paper to *method and synthetic benchmark*; real-data validation (galactic, cosmological) is reported in a companion manuscript.

2 The ADCD framework

Residual extraction. Given a known classical law $y_{\text{classical}} = g(X; \phi)$ and noisy anomalous data y_{obs} , ADCD isolates the correction by forming a residual. In *multiplicative* mode $y_{\text{true}} = y_{\text{classical}}(1 + \Delta)$ gives $\delta = y_{\text{obs}}/y_{\text{classical}} - 1$; in *additive* mode $y_{\text{true}} = y_{\text{classical}} + \Delta$ gives $\delta = y_{\text{obs}} - y_{\text{classical}}$. A mode selector picks whichever best isolates the anomalous deviation.

Proposer. Candidate correction templates $\Delta(X; \theta)$ are drawn from a stochastic sampler over dimensionless ratios (e.g. v/c , r/θ_1), biased by five residual features extracted from δ : monotonicity, curvature sign, oscillation score, exponential decay R^2 , and even/odd symmetry. These features upweight plausible families (e.g. exponentials when monotonicity and decay are high) but never gate candidates—only the physics cascade does.

Stage 1: cascaded physics gates. Every candidate passes three filters *before* optimization:

- G1. AST complexity.** Reject expressions with AST depth > 7 or > 20 tokens—a coarse Occam prior.
- G2. Dimensional homogeneity + transcendental guard.** Verify Δ is dimensionless; physical variables may enter only inside dimensionless ratios. As a sub-check, $\sin, \cos, \exp, \log, \tanh$ may take only *dimensionless* arguments: $\exp(-r)$ without a scale θ_1 is blocked instantly. Free parameters carry adaptive units (relaxable), documented in the companion paper.
- G3. Asymptotic-regime consistency (ARC).** Enforce the decoupling limit: as $x \rightarrow x_0$ (e.g. $v \rightarrow 0$, $r \rightarrow \infty$), $\lim_{x \rightarrow x_0} \Delta(X; \theta) = 0$, evaluated with SymPy’s limit engine.

Survivors are then coarsely ranked by preliminary NMSE on δ , giving an early data-driven signal before the full fit.

Stage 2: JAX-traced optimization + BIC reranking. Surviving expressions are transpiled to JAX and fit by L-BFGS-B (multi-restart, log-uniform init) minimizing NMSE of the residual, $\text{NMSE} = \frac{\sum_i (\Delta(X_i; \theta) - \delta_i)^2}{\sum_i (\delta_i - \delta)^2}$. Under noise, flexible wrong-class expressions can achieve lower NMSE than the truth, so we rerank by the Bayesian Information Criterion [Schwarz \[1978\]](#), [Rissanen \[1978\]](#), $\text{BIC} = N \ln(\text{NMSE}) + k \ln N$, whose $k \ln N$ penalty selects parsimonious correct-class forms. The winner is classified into one of six families (exponential, trigonometric, logarithmic, power law, rational, polynomial) by AST traversal, with degenerate-exponent detection (a fitted $\hat{\theta}_1 \approx 1$ recasts $\theta_0 x^{\theta_1}$ as polynomial).

Table 1: Structural class match rate on the 9-scenario residual benchmark. *generous* doubles PySR’s budget on every axis yet does not close the gap.

Method	0%	1%	5%	10%
ADCD (ours, seed=42)	9/9 (100%)	9/9 (100%)	8/9 (88.9%)	8/9 (88.9%)
ADCD 16-seed mean	—	—	80.4%±7.4%	—
PySR fair	4/9 (44.4%)	5/9 (55.6%)	1/9 (11.1%)	5/9 (55.6%)
PySR generous (2×)	4/9 (44.4%)	4/9 (44.4%)	5/9 (55.6%)	2/9 (22.2%)

Relation to EFT. The cascade mirrors the correction-first philosophy of Effective Field Theory Weinberg [1979]: dimensional analysis of operators (G2), decoupling of heavy modes in the low-energy limit (G3/ARC), and power counting (BIC). ADCD has no renormalization-group flow or controlled E/Λ expansion—it *automates* that philosophy for data-driven problems.

3 Experiments

Benchmark. We evaluate on 9 physical-anomaly scenarios in three tiers: *textbook* (relativistic KE, Yukawa gravity, anharmonic spring), *cross-domain* (net radiation T^{-4} , nonlinear drag, screened Coulomb), and *synthetic novel* formulas. Each is run at 0/1/5/10% observational noise. Success = correct structural *class* recovery (e.g. discovering an exponential correction where the truth is exponential). We report a 16-seed distribution; the mean is the primary claim and seed=42 (the highest seed) is disclosed explicitly, never as the headline.

Baselines. We compare against PySR Cranmer [2020] on the *same residual targets* $\delta = y_{\text{obs}} - y_{\text{classical}}$. The primary baseline is the *fair* profile ($n_{\text{iter}}=100$, $\text{maxsize}=30$, 60 s/scenario). To rule out a budget artefact we add a *generous* profile that *doubles* every budget axis ($n_{\text{iter}}=200$, $\text{maxsize}=40$, 120 s; $\sim 3\times$ wall-clock).

Main result: the gap is structural, not budgetary. Table 1 reports structural match rates. ADCD reaches 88.9% (8/9) at 5% noise versus 11.1% (1/9) for PySR fair—a 77.8 pp gap under matched residuals. The comparison is *seed-independent*: at 5% noise ADCD’s worst of 16 seeds (66.7%, 6/9) still exceeds PySR fair. Crucially, doubling PySR’s budget does *not* close the gap—*generous* total recovery stays flat at 15/36 (identical to *fair*) and at the 5% headline point it reaches only 5/9 (55.6%). The extra budget explores deeper trees (mean complexity 9.8 vs. 8.7 nodes) spent on more deeply-overfit rather than more physically-correct structure. The binding constraint is *structural selection*, not search budget; physics gates resolve it directly.

Why PySR is non-monotonic, and why ADCD is not. PySR fair exhibits a non-monotonic noise profile (4, 5, 1, 5 out of 9). Inspection reveals a mechanistic cause: at 0% noise the large search space lets it find deeply-nested trigonometric expressions (mean 13.8 nodes) that fit clean data but are structurally wrong (5/5 mismatches are trigonometric); at 1% noise regularizes overfitting and complexity drops to 8.2 nodes; at 5% it picks the wrong family; at 10% it collapses to the simplest form, coincidentally correct for 5. ADCD avoids this because its gates are noise-independent: dimensional and ARC constraints reject the nested trig that PySR overfits to, and once the correct family is sampled the cascade selects it regardless of SNR (Section 3).

Ablation: the gates act collectively. Section 3 disables gates one at a time at 5% noise. Removing ARC, AST, or the coarse-eval pre-filter individually leaves accuracy unchanged (8/9); removing the dimensional gate alone costs 11.1 pp (8/9 $\rightarrow$ 7/9). Only disabling *all* gates causes a larger collapse (8/9 $\rightarrow$ 4/9, -44.5 pp): the gates reinforce each other as a collective physical-consistency filter rather than each being independently essential.

4 Discussion and limitations

ADCD is a restriction, not a competitor. ADCD does not compete with PhySO or PySR—it restricts the problem they solve to a physically-motivated subspace where they are known to fail under noise. This makes ADCD directly *combinable* with them: a PhySO/PySR/LaSR Grayeli et al. [2024] proposer could feed the same gate cascade.

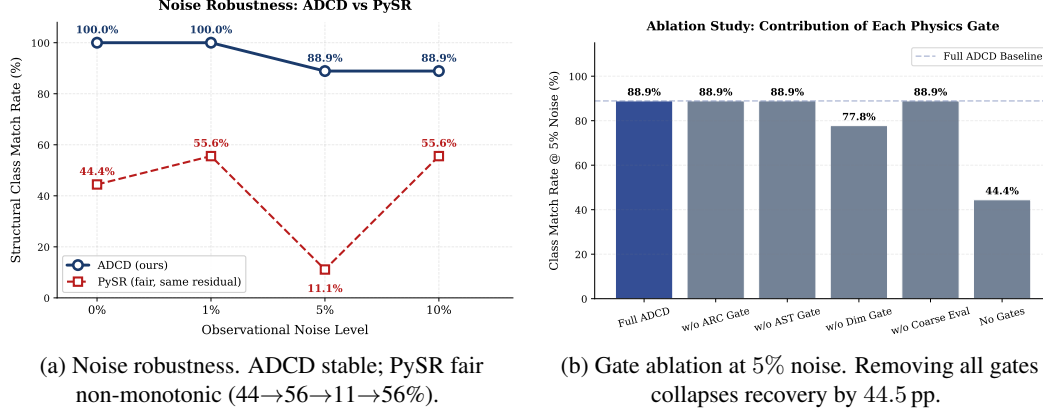


Figure 2: Left: structural match rate vs. noise. Right: per-gate ablation.

Honest limitations. ADCD assumes a structurally-correct classical baseline (incremental correction, not wholesale replacement). At $\geq 5\%$ noise, screened-Coulomb decay is fitted by a rational proxy, reflecting decay-vs-saturation ambiguity under limited dynamic range. A multivariable extension (Buckingham-II) is currently weaker (2/4 at 5%) and treated as exploratory; the dimensional gate uses adaptive parameter-unit relaxation rather than PhySO’s strict enforcement—complementary design points.

Reproducibility. ADCD ships as an open-source Python package (JAX backend, public API/CLI, CI/CD); all 16 seeds, per-seed noise distributions, and PySR profiles are bundled for replication.

Closing. Noisy symbolic regression is bottlenecked by *structural selection*, not search budget. The correction-first formulation—searching for a physically-valid Δ to a known law, screened by dimensional and asymptotic gates before any fit—resolves this directly: ADCD delivers seed-independent gains over PySR, stays robust where unconstrained search is non-monotonic, and is orthogonal to existing SR systems. Future work extends the cascade to multivariable laws and integrates it as a selection layer over external proposers (PhySO, LaSR).

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